step by step:

Git clone:

```
joaoribeiro@JOCAPC:~$ git clone https://github.com/Pokfinner/ironhack-project-1.git
Cloning into 'ironhack-project-1'...
remote: Enumerating objects: 151, done.
remote: Counting objects: 100% (83/83), done.
remote: Compressing objects: 100% (29/29), done.
remote: Total 151 (delta 58), reused 54 (delta 54), pack-reused 68 (from 1)
Receiving objects: 100% (151/151), 137.36 KiB | 1.49 MiB/s, done.
Resolving deltas: 100% (73/73), done.
```

docker build - voting-app

```
joaoribeiro@JOCAPC:~/ironhack-project-1/vote$ docker build -t joaoribeiro9595/voting-app .
[+] Building 11.6s (12/12) FINISHED                                                                    docker:default
 => [internal] load build definition from Dockerfile                                                                0.0s
 => => transferring dockerfile: 1.21kB                                                                              0.0s
 => [internal] load metadata for docker.io/library/python:3.11-slim                                                 1.5s
 => [auth] library/python:pull token for registry-1.docker.io                                                       0.0s
 => [internal] load .dockerignore                                                                                   0.0s
 => => transferring context: 2B                                                                                     0.0s
 => [base 1/5] FROM docker.io/library/python:3.11-slim@sha256:2c078a0e5679ec6484e684c4a8113829dc49c4387f54ad4d04941ee  3.0s
 => => resolve docker.io/library/python:3.11-slim@sha256:2c078a0e5679ec6484e684c4a8113829dc49c4387f54ad4d04941eea8232  0.0s
 => => sha256:2c078a0e5679ec6484e684c4a8113829dc49c4387f54ad4d04941eea82327710 10.37kB / 10.37kB                    0.0s
 => => sha256:ff71127c215572121f1991bacf17f39ec5fcfd2de1f1c01a595835495bb9adfc 1.75kB / 1.75kB                      0.0s
 => => sha256:1eee6fcc4d866c86dd8e32a4a1a583767ddac0d640b0fe571b2c22d233b60eee 5.48kB / 5.48kB                      0.0s
 => => sha256:3531af2bc2a9c8883754652783cf96207d53189db279c9637b7157d034de7ecd 29.78MB / 29.78MB                    1.1s
 => => sha256:91ff8760033c3b9bc9c3a3bcc6ff66c113ec9e13f94fa2f12ae4b221aae996b9 1.29MB / 1.29MB                      1.1s
 => => sha256:f3ba2250c52436e41e9f2f3830af92de466fcacc2d848264e1f9fdc76975d803 14.37MB / 14.37MB                    0.9s
 => => sha256:7ccd73948dde4b7f1623ac6bf9b58706fb83d0a363d765d7c799d46035ceb563 249B / 249B                          1.2s
 => => extracting sha256:3531af2bc2a9c8883754652783cf96207d53189db279c9637b7157d034de7ecd                           0.9s
 => => extracting sha256:91ff8760033c3b9bc9c3a3bcc6ff66c113ec9e13f94fa2f12ae4b221aae996b9                           0.1s
 => => extracting sha256:f3ba2250c52436e41e9f2f3830af92de466fcacc2d848264e1f9fdc76975d803                           0.6s
 => => extracting sha256:7ccd73948dde4b7f1623ac6bf9b58706fb83d0a363d765d7c799d46035ceb563                           0.0s
 => [internal] load build context                                                                                   0.3s
 => => transferring context: 24.54MB                                                                                0.3s
 => [base 2/5] RUN apt-get update &&     apt-get install -y --no-install-recommends curl &&     rm -rf /var/lib/apt/l  4.0s
 => [base 3/5] WORKDIR /usr/local/app                                                                               0.0s
 => [base 4/5] COPY requirements.txt ./requirements.txt                                                             0.0s
 => [base 5/5] RUN pip install --no-cache-dir -r requirements.txt                                                   2.8s
 => [final 1/1] COPY . .                                                                                            0.2s
 => exporting to image                                                                                              0.2s
 => => exporting layers                                                                                             0.1s
 => => writing image sha256:52ea8be9e08ff1bfde934d19307cffbf29aac9e77f583f0541c607e9a9f40d0f                        0.0s
 => => naming to docker.io/joaoribeiro9595/voting-app                                                               0.0s
```

docker push voting-app

```
joaoribeiro@JOCAPC:~/ironhack-project-1/vote$ docker push joaoribeiro9595/voting-app
Using default tag: latest
The push refers to repository [docker.io/joaoribeiro9595/voting-app]
4d602c498727: Pushed
d797f91fb49f: Pushed
d4041a97045d: Pushed
064500430729: Pushed
57cf110034a4: Pushed
2360b28b4660: Mounted from library/python
248c30140986: Mounted from library/python
fa1aec823035: Mounted from library/python
6d7c150df58d: Mounted from library/python
latest: digest: sha256:0e38328539c9139eba4c8cdbef10afd61777b5a3fdc52d1a41e892fe5e7c2401 size: 2206
```

docker build result-app

```
joaoribeiro@JOCAPC:~/ironhack-project-1/result$ docker build -t joaoribeiro9595/result.app .
[+] Building 12.1s (13/13) FINISHED                                                                    docker:default
 => [internal] load build definition from Dockerfile                                                                0.0s
 => => transferring dockerfile: 732B                                                                                0.0s
 => [internal] load metadata for docker.io/library/node:18-slim                                                     1.5s
 => [auth] library/node:pull token for registry-1.docker.io                                                         0.0s
 => [internal] load .dockerignore                                                                                   0.0s
 => => transferring context: 54B                                                                                    0.0s
 => [1/7] FROM docker.io/library/node:18-slim@sha256:f9ab18e354e6855ae56ef2b290dd225c1e51a564f87584b9bd21dd651838830e  3.3s
 => => resolve docker.io/library/node:18-slim@sha256:f9ab18e354e6855ae56ef2b290dd225c1e51a564f87584b9bd21dd651838830e  0.0s
 => => sha256:b98d3ae1ab80d768fc7be41c07c4757d205cc33d2d26261b8273defb45455315 3.31kB / 3.31kB                      0.5s
 => => sha256:b1831021e35a69864dfc4c89eab9ab1232cf3508c787b082c55ee386c5c8527d 38.25MB / 38.25MB                    2.2s
 => => sha256:f9ab18e354e6855ae56ef2b290dd225c1e51a564f87584b9bd21dd651838830e 6.49kB / 6.49kB                      0.0s
 => => sha256:fc3faf127a182135fd956e68d570b1932a758f8008866d8dd6e131cf89de9605 1.93kB / 1.93kB                      0.0s
 => => sha256:101e0128c8ea90af6e5eba2abbae8486503c6383c35cb30e2c60842a5a288479 6.54kB / 6.54kB                      0.0s
 => => sha256:61320b01ae5e0798393ef25f2dc72faf43703e60ba089b07d7170acbabbf8f62 28.23MB / 28.23MB                    0.9s
 => => sha256:c768ab8cba73bb84835e67ed974c93e3e0e3ce1f67a73802422d66021fc9a07a 1.71MB / 1.71MB                      0.9s
 => => sha256:8c994cf49dd19aec88c96926e1a13b36a8f3e159942a5a2d0a9e2dc3e9e6c3dc 448B / 448B                          1.2s
 => => extracting sha256:61320b01ae5e0798393ef25f2dc72faf43703e60ba089b07d7170acbabbf8f62                           1.0s
 => => extracting sha256:b98d3ae1ab80d768fc7be41c07c4757d205cc33d2d26261b8273defb45455315                           0.0s
 => => extracting sha256:b1831021e35a69864dfc4c89eab9ab1232cf3508c787b082c55ee386c5c8527d                           1.0s
 => => extracting sha256:c768ab8cba73bb84835e67ed974c93e3e0e3ce1f67a73802422d66021fc9a07a                           0.0s
 => => extracting sha256:8c994cf49dd19aec88c96926e1a13b36a8f3e159942a5a2d0a9e2dc3e9e6c3dc                           0.0s
 => [internal] load build context                                                                                   0.0s
 => => transferring context: 279.09kB                                                                               0.0s
 => [2/7] RUN apt-get update &&     apt-get install -y --no-install-recommends curl tini &&     rm -rf /var/lib/apt/l  3.4s
 => [3/7] WORKDIR /usr/local/app                                                                                    0.0s
 => [4/7] RUN npm install -g nodemon                                                                                1.5s
 => [5/7] COPY package*.json ./                                                                                     0.0s
 => [6/7] RUN npm ci &&     npm cache clean --force &&     mv /usr/local/app/node_modules /node_modules             1.9s
 => [7/7] COPY . .                                                                                                  0.1s
 => exporting to image                                                                                              0.1s
 => => exporting layers                                                                                             0.1s
 => => writing image sha256:66c15ccea3effdb8b360aa7325077de24e96e0f7a337400a7c1e53bd4c6da7a8                        0.0s
 => => naming to docker.io/joaoribeiro9595/result.app                                                               0.0s

1 warning found (use docker --debug to expand):
- SecretsUsedInArgOrEnv: Do not use ARG or ENV instructions for sensitive data (ENV "PG_PASSWORD") (line 22)
```

docker push result-app

```
joaoribeiro@JOCAPC:~/ironhack-project-1/result$ docker push joaoribeiro9595/result-app
Using default tag: latest
The push refers to repository [docker.io/joaoribeiro9595/result-app]
31ffd66e1f99: Pushed
213142254a54: Pushed
2827fe4736dd: Pushed
aebaf47b47bf: Pushed
750603e3690e: Pushed
9df2f689df70: Pushed
e1e93a4fa4fd: Mounted from library/node
9b95ddc8a140: Mounted from library/node
cd1cd7c2e57b: Mounted from library/node
206f13f5e317: Mounted from library/node
ace34d1d784c: Mounted from library/node
latest: digest: sha256:eaddf66f9f9bcf2b41aa4984cfc36768d32e4f825fa9ee4dbc341f5dd86fa734 size: 2625
```

docker build worker-app

```
joaoribeiro@JOCAPC:~/ironhack-project-1/worker$ docker build -t joaoribeiro9595/worker-app .
[+] Building 26.1s (15/15) FINISHED                                                                  docker:default
 => [internal] load build definition from Dockerfile                                                          0.0s
 => => transferring dockerfile: 706B                                                                          0.0s
 => [internal] load metadata for mcr.microsoft.com/dotnet/runtime:8.0                                         0.8s
 => [internal] load metadata for mcr.microsoft.com/dotnet/sdk:8.0                                             0.8s
 => [internal] load .dockerignore                                                                             0.0s
 => => transferring context: 130B                                                                             0.0s
 => [build 1/6] FROM mcr.microsoft.com/dotnet/sdk:8.0@sha256:4b1cdaa57eed2cecabcf29bdb9bce11e8ca1c287d39dfd2c8b53466  18.8s
 => => resolve mcr.microsoft.com/dotnet/sdk:8.0@sha256:4b1cdaa57eed2cecabcf29bdb9bce11e8ca1c287d39dfd2c8b534663ea94d4  0.0s
 => => sha256:4b1cdaa57eed2cecabcf29bdb9bce11e8ca1c287d39dfd2c8b534663ea94d493 1.08kB / 1.08kB                 0.0s
 => => sha256:1895a935ebe39cbefc5a1b3f5dd9cb50eb3f74b75bc5c03409f6991ac4998601 2.42kB / 2.42kB                 0.0s
 => => sha256:7d09a6b9b026a23e45330fc5d36e2c205e95165e03170e0182901d797cf1ed1e 5.13kB / 5.13kB                 0.0s
 => => sha256:99b64f250d3d876c7921412eeecefc1641821ce569927787df4e9e151fd0d250 3.33kB / 3.33kB                 0.4s
 => => sha256:ff86ea2e5edce334d19a34fbc65d1a511aa1fc823dba1110422f991aa56b44d4 28.24MB / 28.24MB               1.5s
 => => sha256:d120f0decd89c21324dae477e8fa876a9bdf9b1e181e8e81d34464e86c4049f1 18.74MB / 18.74MB               3.2s
 => => sha256:c7e185e5a17988647e52e5183d148a41e455c2bccc2607755bb768c2c784050e 32.25MB / 32.25MB               2.1s
 => => sha256:8c668c44afb8301295862903f1e659df2a04bab6dfc413b3f0be5507dc9217f5 164B / 164B                    1.6s
 => => extracting sha256:ff86ea2e5edce334d19a34fbc65d1a511aa1fc823dba1110422f991aa56b44d4                    23.6s
 => => sha256:f47bbaebda2ea34e23368b75d69b4ddda2ae2351f5a38d878e9bf2dad847543d 11.09MB / 11.09MB               2.5s
 => => sha256:306480c19048ac4ac5f99d0afddb7fd989c455a49abad33be76670dec7599eb6 31.09MB / 31.09MB               3.8s
 => => sha256:40aba635198e120e41d5969de5e30259d9f076e59ea50d258337d4de3763e54b 177.94MB / 177.94MB            15.4s
 => => sha256:b75ce7f87632dd82d3af9169d42f600747546b6c267aa3911d00d58db220474b 2.67MB / 2.67MB                 1.6s
 => => extracting sha256:d120f0decd89c21324dae477e8fa876a9bdf9b1e181e8e81d34464e86c4049f1                    22.0s
 => => extracting sha256:c7e185e5a17988647e52e5183d148a41e455c2bccc2607755bb768c2c784050e                    21.6s
 => => sha256:8013f881b925af18d1418c59c4644ec1d6859d38a53048028efa5f196e2a816b 16.99MB / 16.99MB               4.5s
 => => extracting sha256:8c668c44afb8301295862903f1e659df2a04bab6dfc413b3f0be5507dc9217f5                     0.0s
 => => extracting sha256:f47bbaebda2ea34e23368b75d69b4ddda2ae2351f5a38d878e9bf2dad847543d                     0.1s
 => => extracting sha256:306480c19048ac4ac5f99d0afddb7fd989c455a49abad33be76670dec7599eb6                     0.8s
 => => extracting sha256:40aba635198e120e41d5969de5e30259d9f076e59ea50d258337d4de3763e54b                     3.0s
 => => extracting sha256:b75ce7f87632dd82d3af9169d42f600747546b6c267aa3911d00d58db220474b                     0.0s
 => => extracting sha256:8013f881b925af18d1418c59c4644ec1d6859d38a53048028efa5f196e2a816b                     0.3s
 => [internal] load build context                                                                             0.0s
 => => transferring context: 71.34kB                                                                          0.0s
 => [runtime 1/3] FROM mcr.microsoft.com/dotnet/runtime:8.0@sha256:24117fb11f2f27bdacc6f1ffab16e3a7cddd2417b4eae5e6bc  4.0s
 => => resolve mcr.microsoft.com/dotnet/runtime:8.0@sha256:24117fb11f2f27bdacc6f1ffab16e3a7cddd2417b4eae5e6bc825810a0  0.0s
 => => sha256:99b64f250d3d876c7921412eeecefc1641821ce569927787df4e9e151fd0d250 3.33kB / 3.33kB                 0.4s
 => => sha256:24117fb11f2f27bdacc6f1ffab16e3a7cddd2417b4eae5e6bc825810a0d1746d 1.08kB / 1.08kB                 0.0s
 => => sha256:9d47ce787b8e1d50dc74a2cba4cf8162c39e1300b98fab7392e437312756dcb6 1.37kB / 1.37kB                 0.0s
 => => sha256:664bf98a9719a0976be4cd9a9ab87affe818ee3b6145632f99aed1fe695c0685 2.17kB / 2.17kB                 0.0s
 => => sha256:ff86ea2e5edce334d19a34fbc65d1a511aa1fc823dba1110422f991aa56b44d4 28.24MB / 28.24MB               1.5s
 => => sha256:d120f0decd89c21324dae477e8fa876a9bdf9b1e181e8e81d34464e86c4049f1 18.74MB / 18.74MB               3.2s
 => => sha256:c7e185e5a17988647e52e5183d148a41e455c2bccc2607755bb768c2c784050e 32.25MB / 32.25MB               2.1s
 => => extracting sha256:ff86ea2e5edce334d19a34fbc65d1a511aa1fc823dba1110422f991aa56b44d4                     1.1s
 => => sha256:8c668c44afb8301295862903f1e659df2a04bab6dfc413b3f0be5507dc9217f5 164B / 164B                    1.6s
 => => extracting sha256:d120f0decd89c21324dae477e8fa876a9bdf9b1e181e8e81d34464e86c4049f1                     0.3s
 => => extracting sha256:99b64f250d3d876c7921412eeecefc1641821ce569927787df4e9e151fd0d250                     0.0s
 => => extracting sha256:c7e185e5a17988647e52e5183d148a41e455c2bccc2607755bb768c2c784050e                     0.3s
 => => extracting sha256:8c668c44afb8301295862903f1e659df2a04bab6dfc413b3f0be5507dc9217f5                    21.3s
 => [runtime 2/3] WORKDIR /app                                                                                0.2s
 => [build 2/6] WORKDIR /src                                                                                  0.2s
 => [build 3/6] COPY Worker.csproj .                                                                          0.0s
 => [build 4/6] RUN dotnet restore                                                                            3.3s
 => [build 5/6] COPY . .                                                                                      0.0s
 => [build 6/6] RUN dotnet publish -c Release -o /app/publish                                                 2.8s
 => [runtime 3/3] COPY --from=build /app/publish ./                                                           0.0s
 => exporting to image                                                                                        0.0s
 => => exporting layers                                                                                       0.0s
 => => writing image sha256:248ab34639b1f49a2fead417533a52245c091ed9875adc12325b8c497abe31a8                  0.0s
 => => naming to docker.io/joaoribeiro9595/worker-app                                                         0.0s

1 warning found (use docker --debug to expand):
- SecretsUsedInArgOrEnv: Do not use ARG or ENV instructions for sensitive data (ENV "DB_PASSWORD") (line 23)
```

docker push result-app

```
joaoribeiro@JOCAPC:~/ironhack-project-1/worker$ docker push joaoribeiro9595/worker-app
Using default tag: latest
The push refers to repository [docker.io/joaoribeiro9595/worker-app]

bbf9eb799c79: Pushed
f945b30421d3: Pushed
0073e24979ab: Pushed
59c5ef0c9efe: Pushed
1a5a51bd93a7: Pushed
9cf6ea2f556f: Pushed
0da811fd3ed4: Pushed
latest: digest: sha256:87dc40babce9d5796cad8c8a2f15d64fac994557bc592e6ec5b9bd72d954de34 size: 1785
```

docker networks creations

```
joaoribeiro@JOCAPC:~/ironhack-project-1/worker$ docker network create frontend
ce2e41e449d14a9012097476a8436a073221ffd251447a77ca4350672bc7a882
joaoribeiro@JOCAPC:~/ironhack-project-1/worker$ docker network create backend
7be8685c2417a9edb7ddd04acc6ce1b38b92bcb0eb984b5575a7cc412189cd06
```

docker run voting-app

```
joaoribeiro@JOCAPC:~/ironhack-project-1/worker$ docker run -d --name voting-app --network frontend -p 5000:80 joaoribeiro9595/voting-app
d955550d49d9790c0ccd30cf26e3efafc3a8d897c9edd02832dbe58040bc48c2
```

docker run redis

```
joaoribeiro@JOCAPC:~/ironhack-project-1/worker$ docker run -d --name redis --network frontend redis:latest
Unable to find image 'redis:latest' locally
latest: Pulling from library/redis
3531af2bc2a9: Already exists
97afe6025aa9: Pull complete
9e0f42ff49e1: Pull complete
d55df8e0f53d: Pull complete
14d3b17b0010: Pull complete
4f4fb700ef54: Pull complete
2b8951cd65b5: Pull complete
Digest: sha256:d372cf7cd5ab47fb6ad1a73c45ea1104d6f3fa11cc833ff0b3ac997890b4ccec
Status: Downloaded newer image for redis:latest
cf27b42e6f03a888c8122ceab193df1f75aef94320e7b8fa54ca1f0e2c977d5f
```

docker run postgres

```
joaoribeiro@JOCAPC:~/ironhack-project-1/worker$ docker run -d --name db_postgres --network backend -e POSTGRES_USER=postgres
_joao -e POSTGRES_PASSWORD=postgres_joao postgres:latest
Unable to find image 'postgres:latest' locally
latest: Pulling from library/postgres
3531af2bc2a9: Already exists
4e2103318c3c: Pull complete
13f7f91984c8: Pull complete
e9cad8a9e695: Pull complete
56b9e522a2ac: Pull complete
c3f159f3b349: Pull complete
6fadfe1078a3: Pull complete
85f6ddcd85b0: Pull complete
c43308a19dbe: Pull complete
2428a49310a9: Pull complete
ce10992ea794: Pull complete
b7036600a44b: Pull complete
e57497b6cf23: Pull complete
Digest: sha256:8da22df20409c62981f57682595ef617ee365ac905c48ab61d0ff8aafa83891d
Status: Downloaded newer image for postgres:latest
18ee15e071deecf572d13efd576bceffd463259d6a80044712cdeda3a0c311be
```

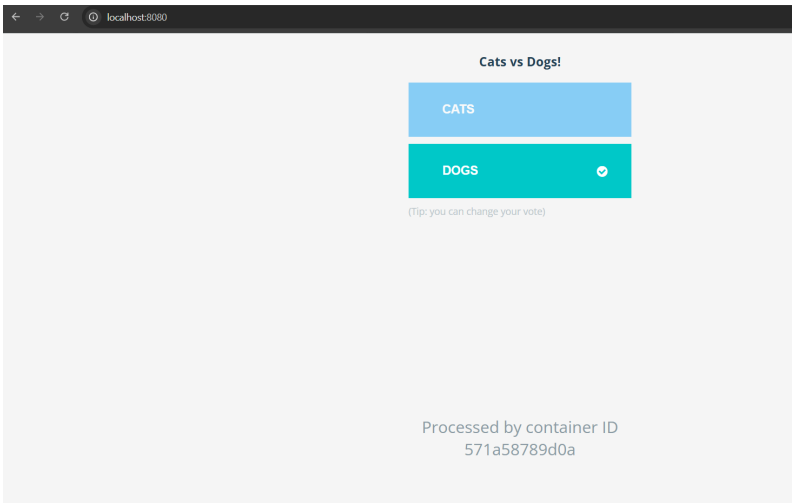
docker run worker-app

```
joaoribeiro@JOCAPC:~/ironhack-project-1/worker$ docker run -d --name worker-app --network frontend --network backend joaoribeiro9595/worker-app
33de7a8e3c1a7d94b4565e88d43a56d2e060cd246c9bbfc35213178ef67a2b0a
```

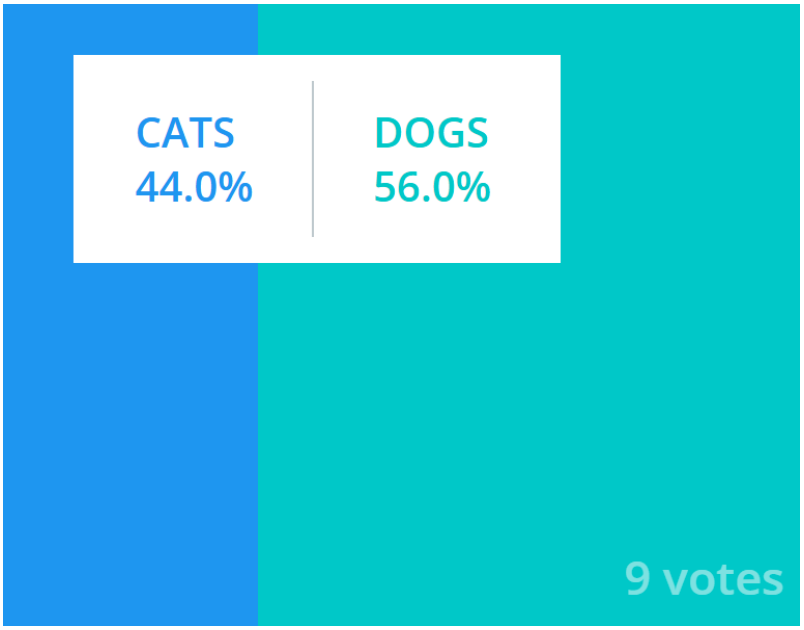
docker run worker-app

```
joaoribeiro@JOCAPC:~/ironhack-project-1/worker$ docker run -d --name result-app --network backend -p 5001:80 joaoribeiro9595/result-app
7b00a886377ad0d6013494feee5759447aa575a5f32eab8fb3f6fbaffe9a4ad9
```

localhost:8080



localhost:8082



NOTE: in the middle of the LAB, i´ve changed de ports. **5000 -> 8080 and 5001 -> 8082**